Evaluating AI Agents

Monitoring Agents





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Talking Head

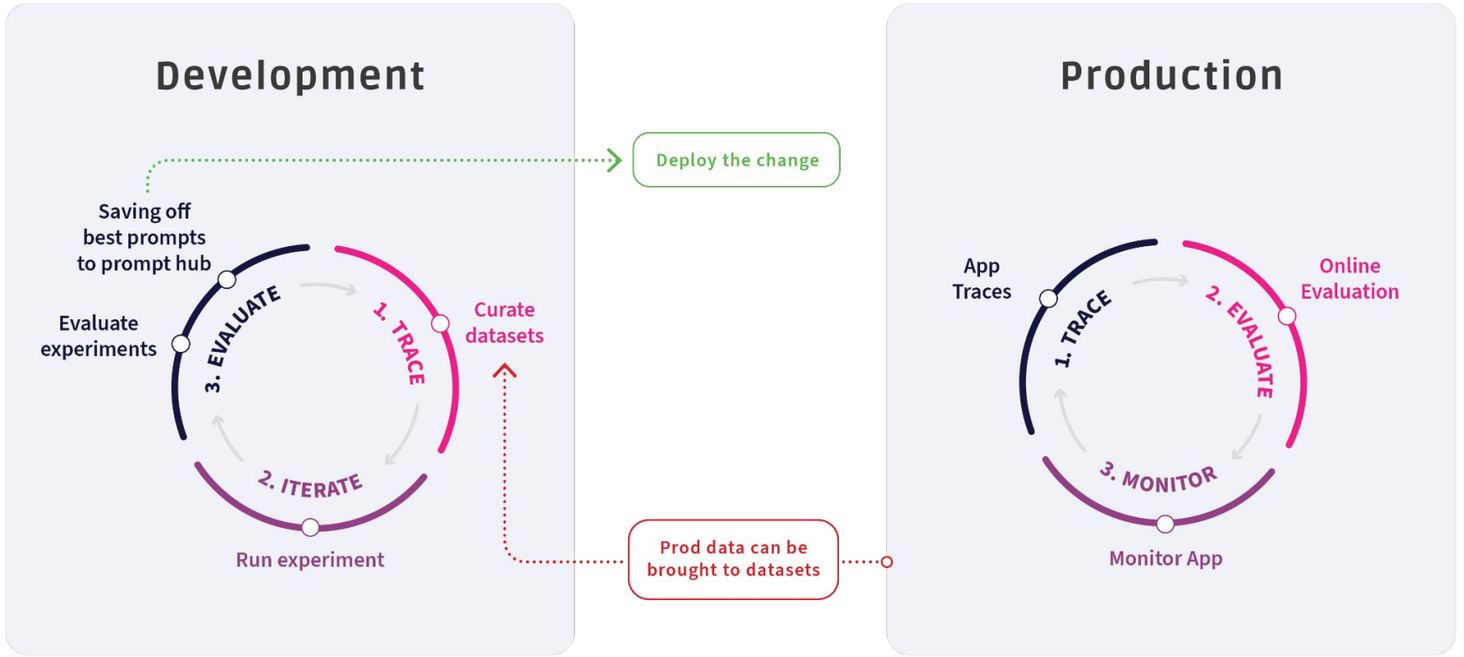
In this lesson:

- Production agents
- Why monitoring agents matters
- Recap: Instrumentation, evals and metrics over time
- Using data from production to iterate on your agent and evals (Adding experiments to a CI/CD pipeline)

Recap: Going from Development to Production

4 Steps to go from development to production

1. Choose the right agent architecture
2. Choose **what** to evaluate and the correct **metrics**
3. Build your eval structure
4. Use your data to iterate

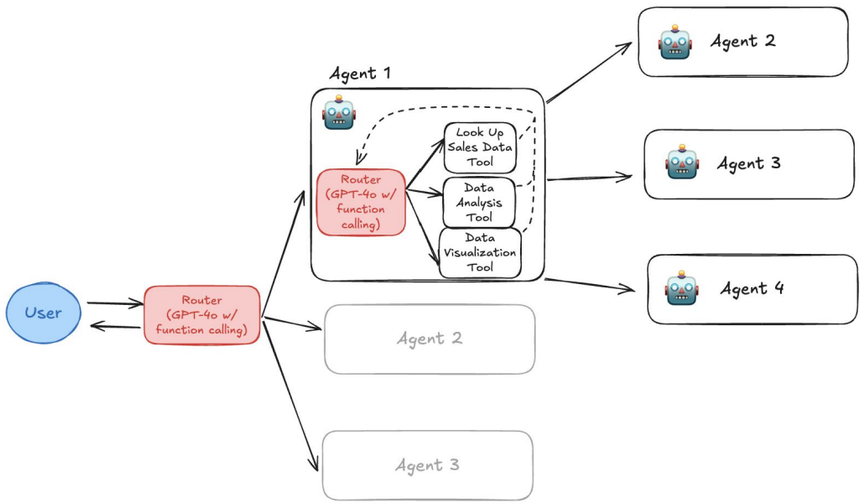


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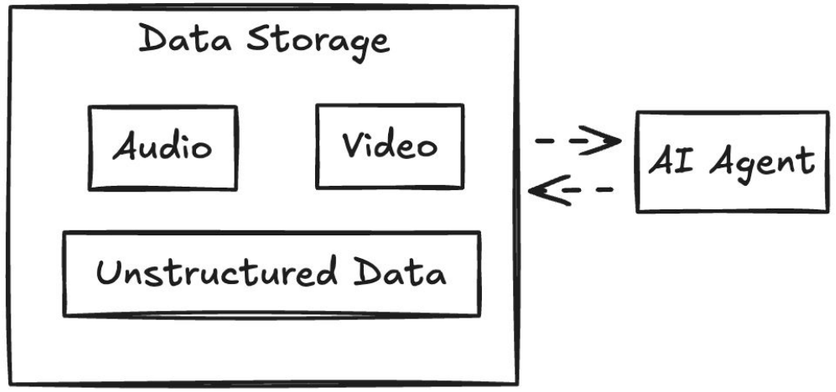
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Scaling Production Agent Systems

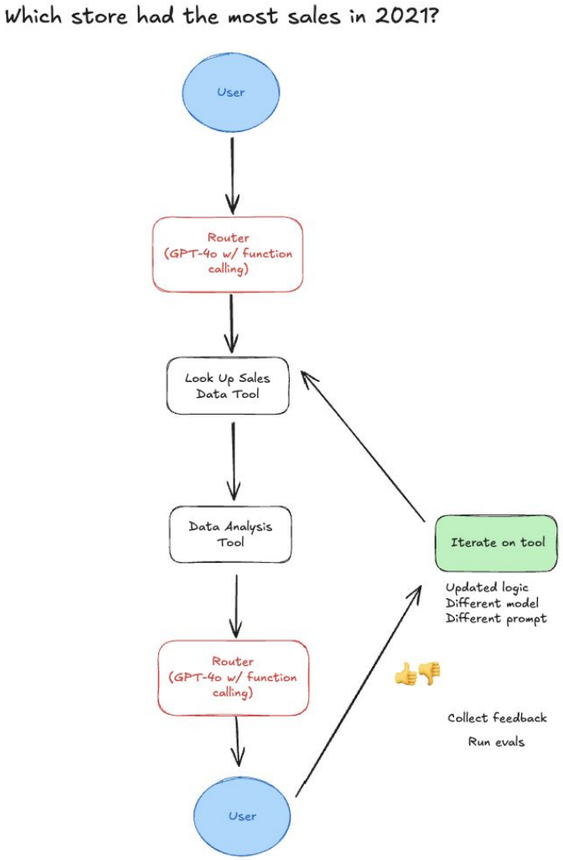
Multi agent systems



Multi-modal systems



Continuously improving agents



What's different about production?

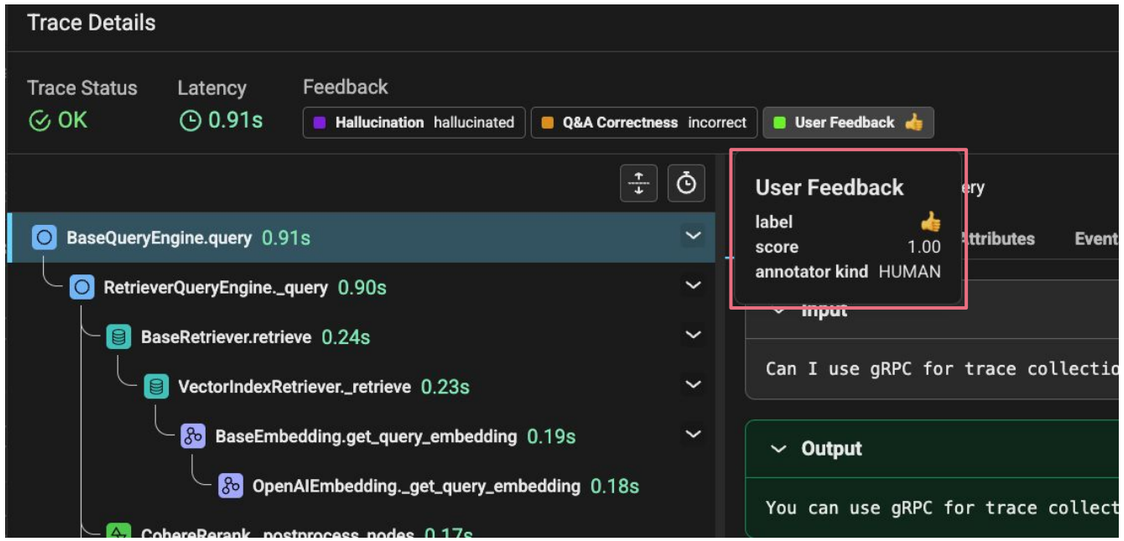
- Discovering new failure modes
  - i.e. new queries or use cases
    - A new product releases that the system is not aware of
- Higher complexity
  - external APIs and other agents
- Making changes to agents
  - a/b testing and evaluating, using data

Tools for managing agent performance in production

- Instrumentation and feedback
- Monitoring your metrics
- CI/CD (continuous improvement)

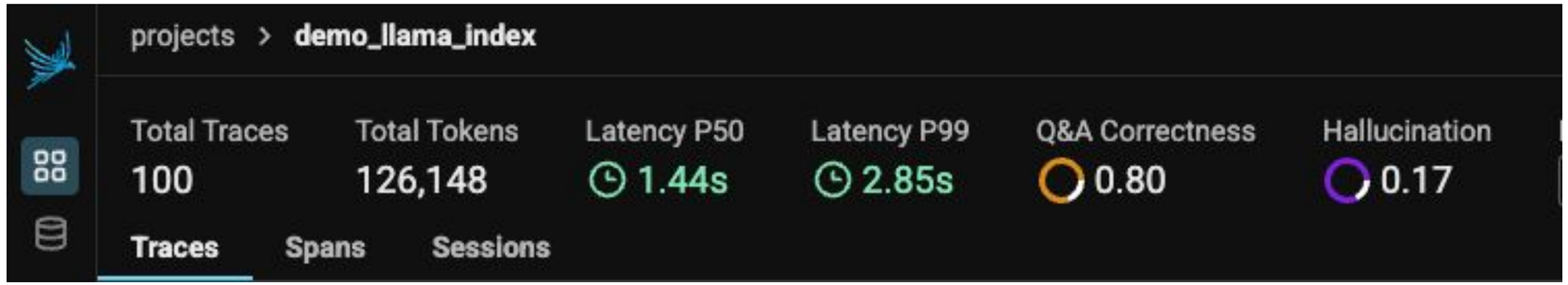
Instrumentation and feedback

- Collect user feedback and data for evals from your application → **Tracing and Annotations**
- Define areas of improvement by looking at your data



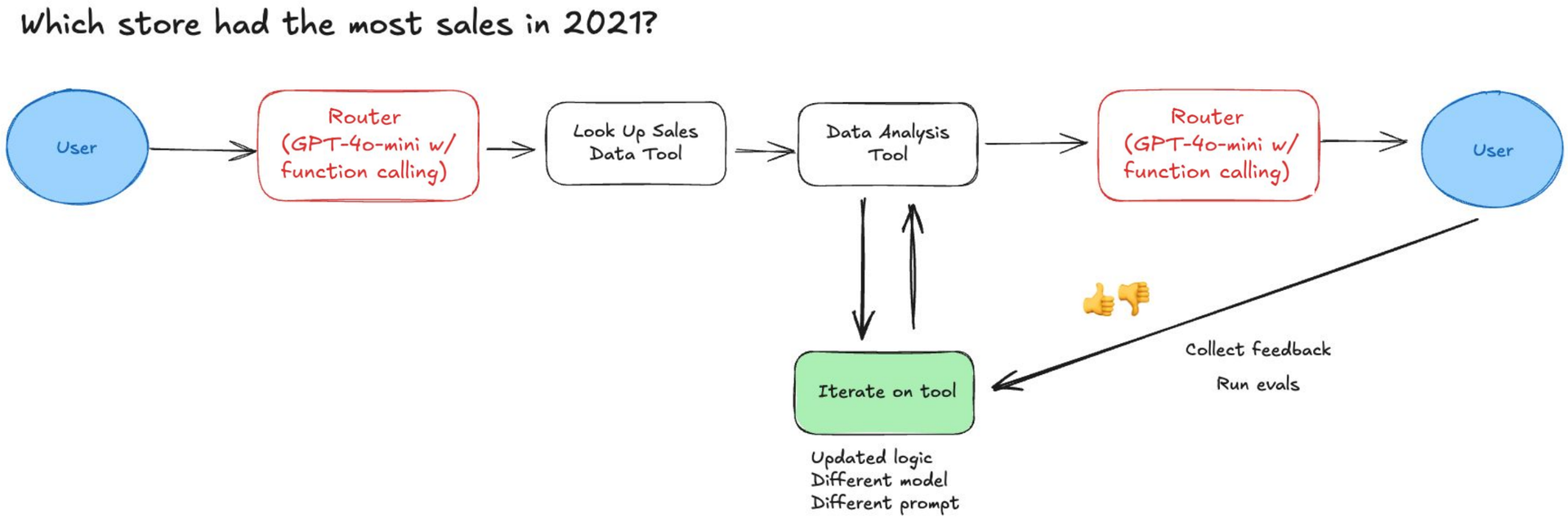
Monitoring metrics

- Efficiency of your system (i.e. Convergence evals)
  - Ensure your system performance does not degrade over time
- Dependencies on external services (latency, cost)
  - Ensure that you are monitoring external LLM calls as they can have a large impact on end user experience
    - i.e. reasoning model vs smaller model for a certain application



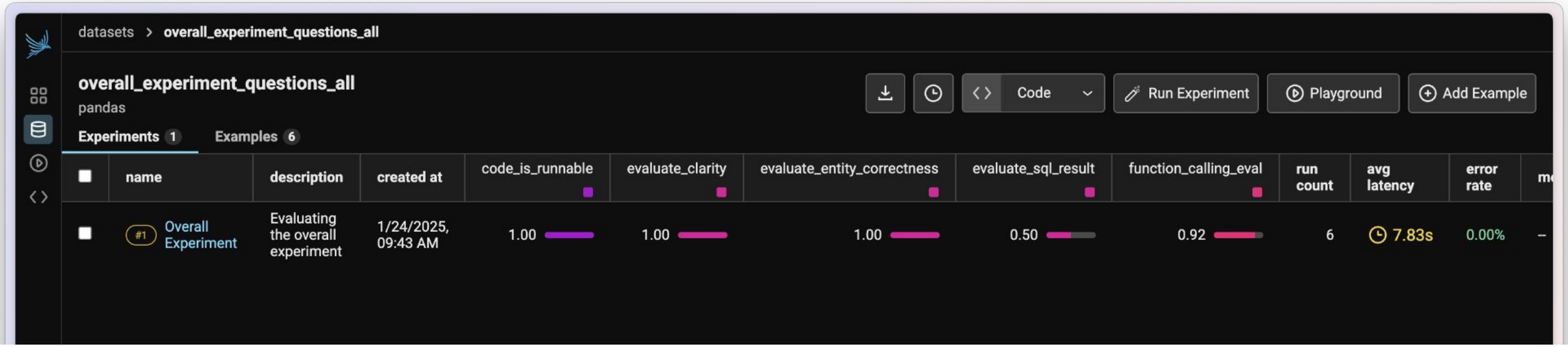
CI/CD (continuous improvement)

- Collect user feedback (human labels) to understand end-to-end Agent performance

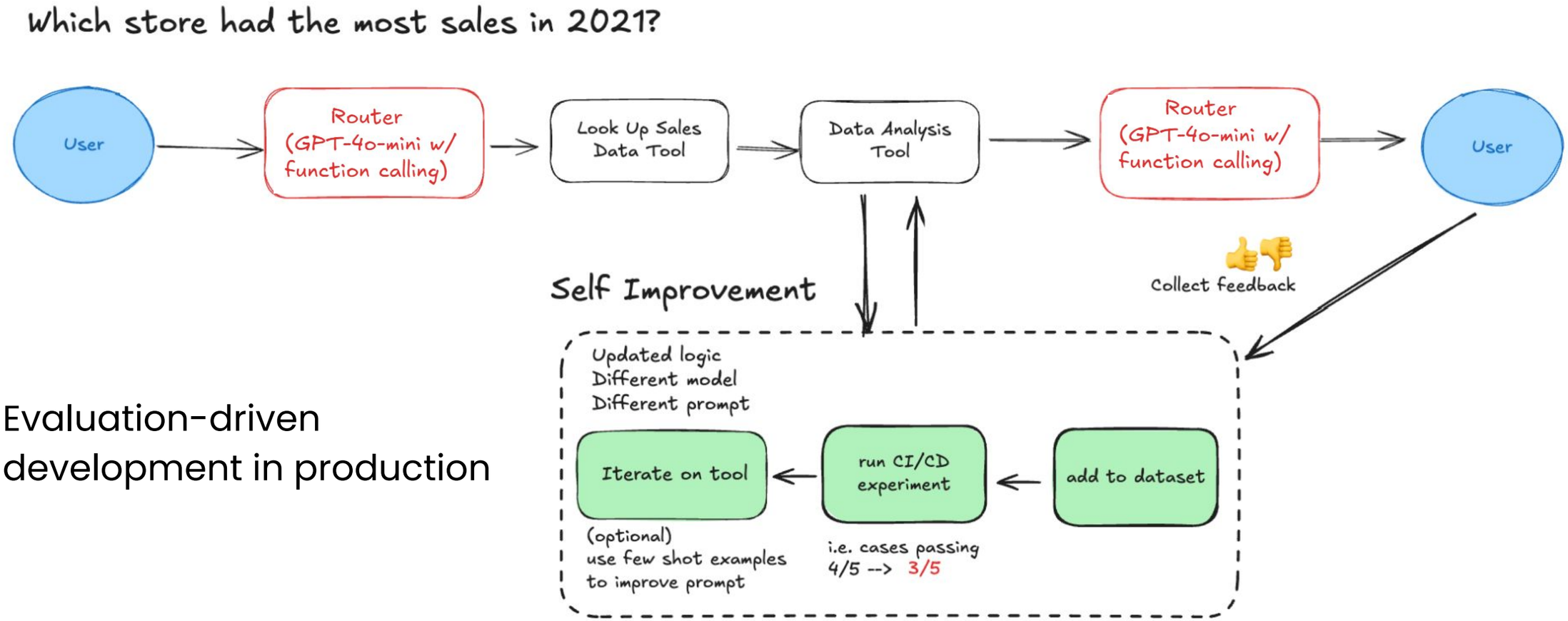


CI/CD (continuous improvement)

- Curate datasets for continuous experimentation as you iterate
  - Datasets can be representative of specific failure modes + general performance (golden datasets)
  - Experiments act as “gates” for shipping changes



Example: Self Improving Agent



Evaluation-driven development in production

What you've learned:

1. What to watch out for when your agent is in production
2. How to use tooling and data from development through to production
3. Future architectural considerations for CI/CD